

Problem 2(a): Scaling law reliability (d4-d9)

Width grows with depth (hidden = 64 x depth). All models: 614M tokens (600k seq x 1024), bs 64, warmup 1%, wd 0.1, AdamW (0.9, 0.95). Fits use d4-d7 only; d8 and d9 have not been trained with these settings yet (except the d8 baseline recipe, run earlier: 2.927).

Final validation loss

Variant	d4 (5.9M)	d5 (10.0M)	d6 (16.0M)	d7 (24.0M)
Baseline (linear, LR 3e-3)	3.290	3.153	3.057	2.983
Constant LR	3.421	3.330	3.243	3.218
Dropout 0.2	3.452	3.325	3.216	3.129
LR 0.03	3.321	3.203	3.105	3.078

Fit 1: power law $L = A * N^{-\alpha}$

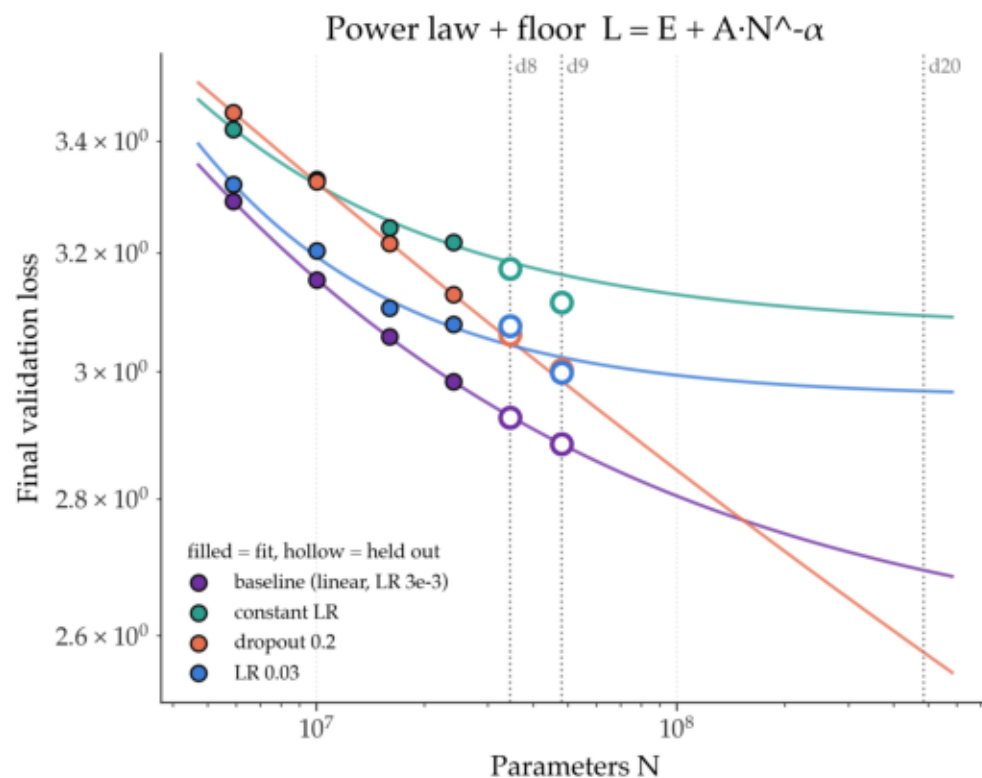
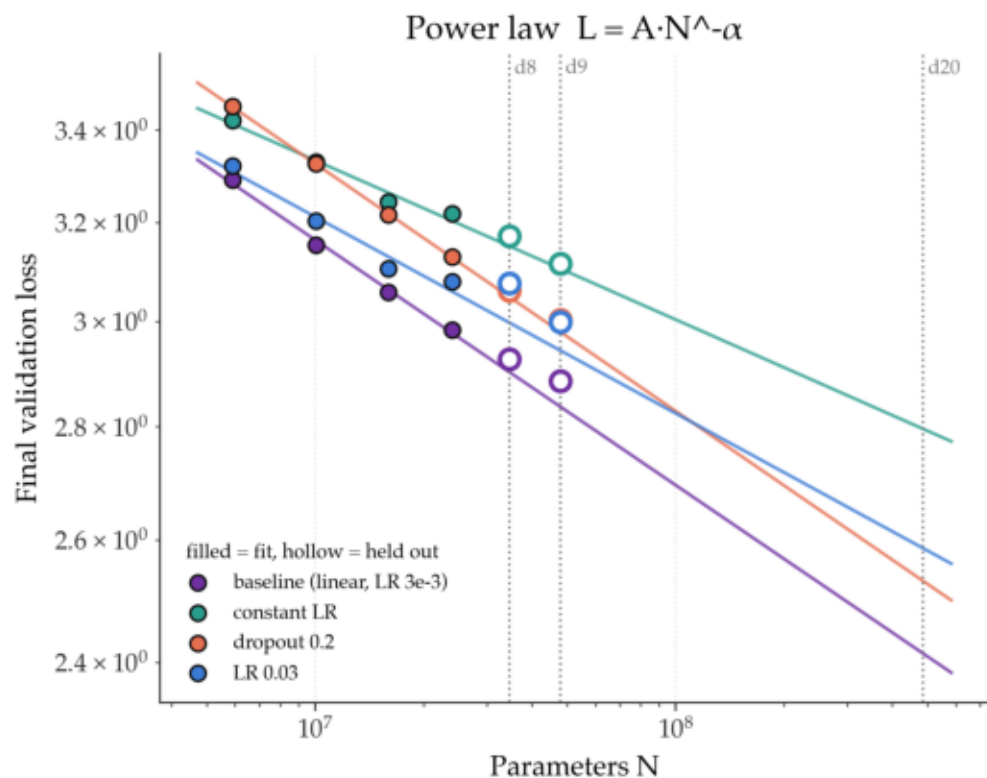
Variant	alpha	d8 (34.6M)	d9 (48.0M)	d20 (486M)
Baseline	0.070	2.902	2.836	2.414
Constant LR	0.045	3.151	3.105	2.796
Dropout 0.2	0.070	3.048	2.979	2.532
LR 0.03	0.056	2.997	2.943	2.586

Fit 2: power law + floor $L = E + A * N^{-\alpha}$

Variant	alpha	E (floor)	d8	d9	d20
Baseline	0.389	2.56	2.928	2.885	2.694
Constant LR	0.648	3.07	3.184	3.162	3.092
Dropout 0.2	0.106	1.10*	3.051	2.985	2.576
LR 0.03	0.822	2.96	3.044	3.024	2.969

* Dropout's points are still nearly straight on log-log axes, so its floor is not identifiable; prefer fit 1 for dropout. The fits agree closely at d8/d9 but diverge at d20: with fixed data (~1.3 tokens/param at d20), a floor is expected.

Fits on d4-d7 (filled), held-out d8/d9 (hollow)



Left: pure power law (straight lines on log-log axes). Right: power law with an irreducible-loss floor E . Constant LR and LR 0.03 are already flattening at d7; the baseline bends gently; dropout is still nearly straight.

Pre-registered predictions (made BEFORE training d8 and d9)

Recorded 2026-09-26, before stage 2 (d8, d9) was launched. * d8 baseline recipe was already known (2.927).

Variant	Fit used	d8 prediction	d9 prediction	d20 prediction	Reasoning
Baseline (linear, LR 3e-3)	floor	2.93 (2.92-2.94)*	2.88 (2.87-2.90)	2.70 (2.62-2.76)	data-limited; floor fit matched known d8
Constant LR	between	3.17 (3.15-3.19)	3.14 (3.11-3.17)	2.95 (2.85-3.10)	no annealing; flattening but d7 step near noise
Dropout 0.2	power law	3.05 (3.03-3.07)	2.99 (2.97-3.01)	2.79 (2.72-2.87)	single epoch: dropout is pure cost; stays > baseline
LR 0.03	floor	3.04 (3.00-3.10)	3.03 (2.98-3.12)	>= 3.0 / unstable	gap to baseline grows with width (0.03 -> 0.10)

How the predictions were made

1. For each variant, look at the plot: are the 4 points a straight line on log-log axes (use fit 1) or bending flat (use fit 2)?
2. Take d8/d9 from the chosen fit. The two fits differ by < 0.05 there, so this choice matters little.
3. For d20, reason about data: every model sees the same 614M tokens, so large models become data-limited and loss should level off. A pure power law is likely too optimistic; the floor fit (or something between the two) is safer.
4. Adjust by judgment if a fit looks unreasonable (e.g. dropout's floor E = 1.10), and write one line of reasoning.
5. Optionally give a range (e.g. d9 baseline 2.87-2.90) - it makes the check after stage 2 more informative.
6. Then run stage 2 (d8, d9) and compare: which fit type predicted better, and does that change your d20 answer?

Stage 2 results: predictions vs actual (d8, d9)

Variant	d8 predicted	d8 actual	d9 predicted	d9 actual	Better raw fit ($ \text{err} $ d8 / d9)
Baseline	2.93 (2.92-2.94)*	2.926	2.88 (2.87-2.90)	2.885	floor (0.002 / 0.000)
Constant LR	3.17 (3.15-3.19)	3.172	3.14 (3.11-3.17)	3.115	power law (0.021 / 0.010)
Dropout 0.2	3.05 (3.03-3.07)	3.061	2.99 (2.97-3.01)	3.003	floor (0.010 / 0.018)
LR 0.03	3.04 (3.00-3.10)	3.075	3.03 (2.98-3.12)	2.999	floor (0.031 / 0.025)

All 8 held-out results fell inside the pre-registered ranges. * d8 baseline was not blind.

Final d20 predictions (refit on all of d4-d9)

Variant	d20 prediction	Why
Baseline	2.69 (2.62-2.75)	floor fit on 6 points: RMSE 0.0008, E = 2.56 stable; PL says 2.48
Constant LR	2.94 (2.85-3.00)	d7 dip was noise: d8/d9 kept improving $\sim 0.05/\text{step}$; floor E = 2.71
Dropout 0.2	2.74 (2.68-2.80)	gap to baseline shrinks 0.16 $\rightarrow$ 0.12; stays just above baseline
LR 0.03	2.93 (2.85-3.05)	non-monotone (d7 3.078, d8 3.075, d9 2.999): noisy, instability risk

- Fixed data (614M tokens, ~ 1.3 tokens/param at d20) makes large models data-limited: the floor model predicted d8/d9 best.
- Pure power laws were too optimistic for the baseline (0.02-0.05 too low by d9) and would be far too optimistic at d20.
- Ranking at d20: baseline < dropout < LR 0.03 $\sim$ constant. Decay and a sane LR matter more at scale; dropout's cost shrinks.
- Caveat: 6 points spanning 8x in parameters, extrapolated 10x further; a single seed (noise ~ 0.006). Treat d20 as ± 0.05 .